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1 # NAME: DAMILOLA GRACE AYOBAMI
2 # MATRIC NO: RSG/20/5390
3 # DEPARTMENT: REMOTE SENSING AND GEOSCIENCE INFORMATION SYSTEM
4 # -----
5
6 # Constants
7 daily_extraction = 1200
8 selling_price = 20000
9 processing_cost_per_ton = 6000
10 fixed_cost = 10000000
11 royalty_rate = 0.05
12
13 # Efficiencies for 7 days
14 efficiencies = [0.75, 0.78, 0.8, 0.79, 0.73, 0.74, 0.76]
15
16 # Step 1: Calculate daily effective outputs
17 daily_outputs = []
18
19 for eff in efficiencies:
20     output = daily_extraction * eff
21     daily_outputs.append(output)
22
23 # Step 2: Total effective output
24 total_effective_output = sum(daily_outputs)
25
26 # Step 3: Total extracted (before efficiency loss)
27 total_extracted = daily_extraction * 7
28
29 # Step 4: Revenue
30 total_revenue = total_effective_output * selling_price
31
32 # Step 5: Processing cost
33 total_processing_cost = total_extracted * processing_cost_per_ton
34
35 # Step 6: Royalty
36 royalty = royalty_rate * total_revenue
37
38 # Step 7: Total cost
39 total_cost = total_processing_cost + fixed_cost + royalty
40
41 # Step 8: Net profit
42 net_profit = total_revenue - total_cost
43
44 # OUTPUTS
45 print("Daily Effective Outputs:")
46 for i, val in enumerate(daily_outputs, start=1):
47     print(f"Day {i}: {val:.2f} tons")
48
49 print("\nTotal Effective Output:", total_effective_output)
50 print("Total Revenue: ₦", total_revenue)
51 print("Processing Cost: ₦", total_processing_cost)
52 print("Royalty: ₦", royalty)
53 print("Total Cost: ₦", total_cost)
54 print("Net Profit: ₦", net_profit)
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